

INVENTORY MANAGEMENT SYSTEM

Prepared By: **CSD026**

SOUMYA ARORA	2316572
SUHANI JAIN	2316576
SOMYA JAIN	2316571
PALAK PATODI	2316496
VANSHREE	2316602



Introduction

Problem Statement

- The Faculty of Mathematics and Computing is currently managing and tracking inventory such as stationery, furniture, computer sets, and miscellaneous items.
- Manual record-keeping leads to errors, inefficiency, and difficulty in monitoring stock availability.
- No centralized system exists for allocation, tracking, and updating inventory usage.
- Limited visibility for the HOD and staff makes accountability and timely decision-making difficult.

Solution

- Develop a desktop-based Inventory Management Application (IMA) tailored for departmental needs.
- Provide authorized access for staff (day-to-day operations) and the HOD (administrative control).
- Streamline processes for:
 - Record-keeping (add, update, delete inventory items)
 - Item allocation and tracking
 - Monitoring stock levels in real time
 - Generate comprehensive reports (e.g., stock summary, usage history, budget analysis) for better decision-making.
- Ensure accuracy, efficiency, and transparency in inventory management.

01

Track real-time availability of products, avoiding stockouts or overstocking.

02

Ensure materials/products are available when needed to meet production requirements.

03

Streamline purchasing, storage, and distribution processes.

04

Maintain clear records of stock movement, ownership, and usage.

05

Automated calculations minimize human errors in stock counts and data entry.

06

Digital records are searchable; no need to shuffle through piles of registers.

07

Reduced paper, storage, and administrative costs; less wastage due to better control.

08

Users can access analytics, and reports instantly.

09

As inventory grows, digital systems handle large volumes easily without proportional manpower increase.

10

Easier reporting and accurate logs simplify internal and external audits.

PRODUCT SCOPE

FUNCTIONS

- **Authentication & Access Control**

Secure logins for Inventory Manager, HOD and Dean

- **Product Management**

Product & Product Type tables for stationery, furniture, computer sets, etc.; track vendor/purchase records, staff/room allocations, usage & availability

- **Procurement & Suppliers**

Supplier & Bill tables for vendor details and purchase tracking

- **Issuance & Returns**

Issue & Return tables for items issued/returned; maintain accurate stock levels & history

FUNCTIONS

- **Item Request Management**

Inventory Manager raise requests for new/replenishment items via mail

- **Reporting & Monitoring**

Summaries, usage analysis, availability checks, budget analysis; reports with dynamic graphs & visualizations

- **Automation**

Replace manual, paper-based processes; reduce errors, improve access

- **Oversight**

HOD & Dean approvals, high-level reporting & rights; mandatory approvals for large/computer orders; ensures transparency & efficiency

Features

1.Login

- **Description & Priority**

Manages secure login for Inventory Managers, HOD and the Dean.

High priority: protects sensitive data, enforces user privileges.

- **Stimulus/Response**

User enters username & password.

If valid → access granted as per role.

If invalid → error message, re-enter credentials.

Includes Forgot Password link to reset password.

- **Functional Requirements**

1.REQ-1: Valid username & password required.

2.REQ-2: Restrict access until authentication.

3.REQ-3: Support three roles (Manager, HOD, Dean).

4.REQ-4: Role-based interface & features.

5.REQ-5: Lockout after 3 failed attempts.

6.REQ-6: Authenticated users can change password.



Features

2.Product Management

- **Description & Priority**

Manages product records, purchases, and allocations.

Uses common database with ProductType Table.

High priority: tracks inventory from procurement to assignment.

- **Stimulus/Response**

Manager opens Product Mgmt. module.

Add supplier records, purchase bills, item issuance.

Input purchase details (supplier, bill, date, type, quantity).

Raise request for new/replenished items → sent to HOD/Dean.

Record allocations (staff, department, room, quantity).

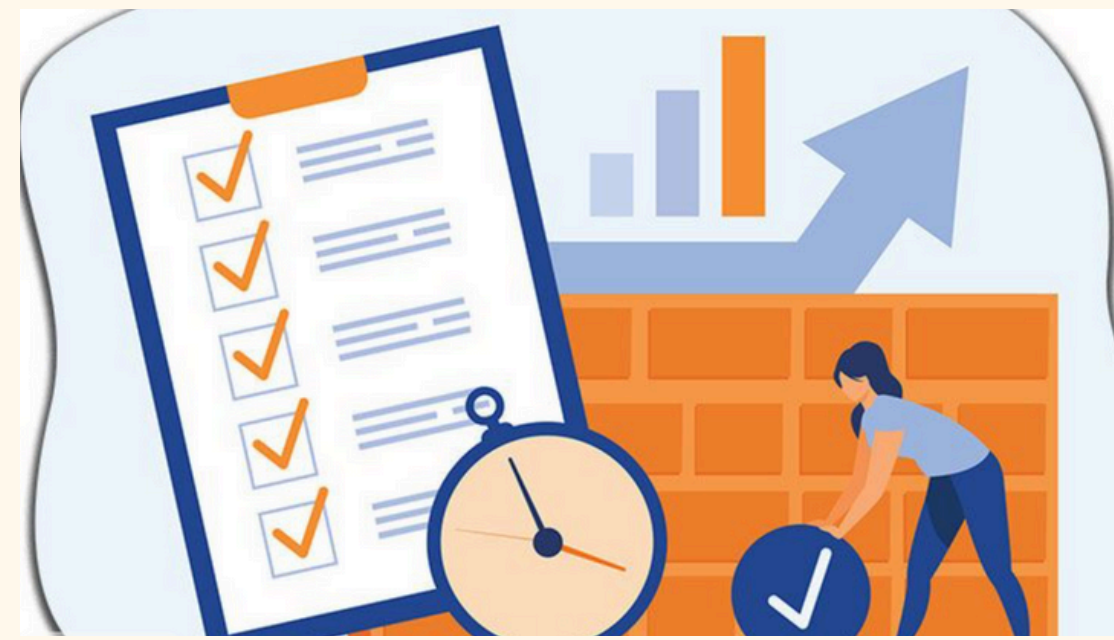
Validate & save data → update Product & Issue tables.

Record returns → update stock.

Success confirmation or error highlights.



Features



2.Product Management

- **Functional Requirements**

REQ-7: CRUD Supplier & Bill records.

REQ-8: CRUD issuance & return records.

REQ-9: Validate all inputs (date, quantity).

REQ-10: Prevent issuing more than stock.

REQ-11: Search/filter on all tables.

REQ-12: Invalid inputs → error messages.

REQ-13: Dedicated return function → update stock.

REQ-14: Managers raise requests → needs admin approval.



Features



3.Reporting & Administrative Overview

- **Description & Priority**

Provides reporting & monitoring, role-based visibility.
High priority: supports decision-making & transparency.

- **Stimulus/Response**

User → Reports section → dashboard shown.

Inventory Managers: Auto daily/monthly/yearly reports (stock, allocations, issues).

HOD: Budget Analysis & Faculty Usage reports.

Dean: Access all reports.

Export to PDF.

Validate reports with database.



Features



3.Reporting & Administrative Overview

- **Functional Requirements**

REQ-15: Real-time reports for Managers, HOD, Dean.

REQ-16: Export reports in PDF.

REQ-17: Search & filter reports.

REQ-18: Reporting interface only for Admin roles.

REQ-19: Log report generation for audit.

REQ-20: Auto daily/monthly/yearly reports for Managers.

REQ-21: Monthly summary available at start of each month.



01

Operating System: Windows 10 or higher (Linux optional with Java setup).

02

Frontend: JavaFX-based GUI.
Backend: Java-based logic with SQL database.

03

Runtime: Java SE version 17 LTS or higher.

04

Hardware: Standard departmental PCs (at least 4 GB RAM or higher as per data, Mid-range CPU).

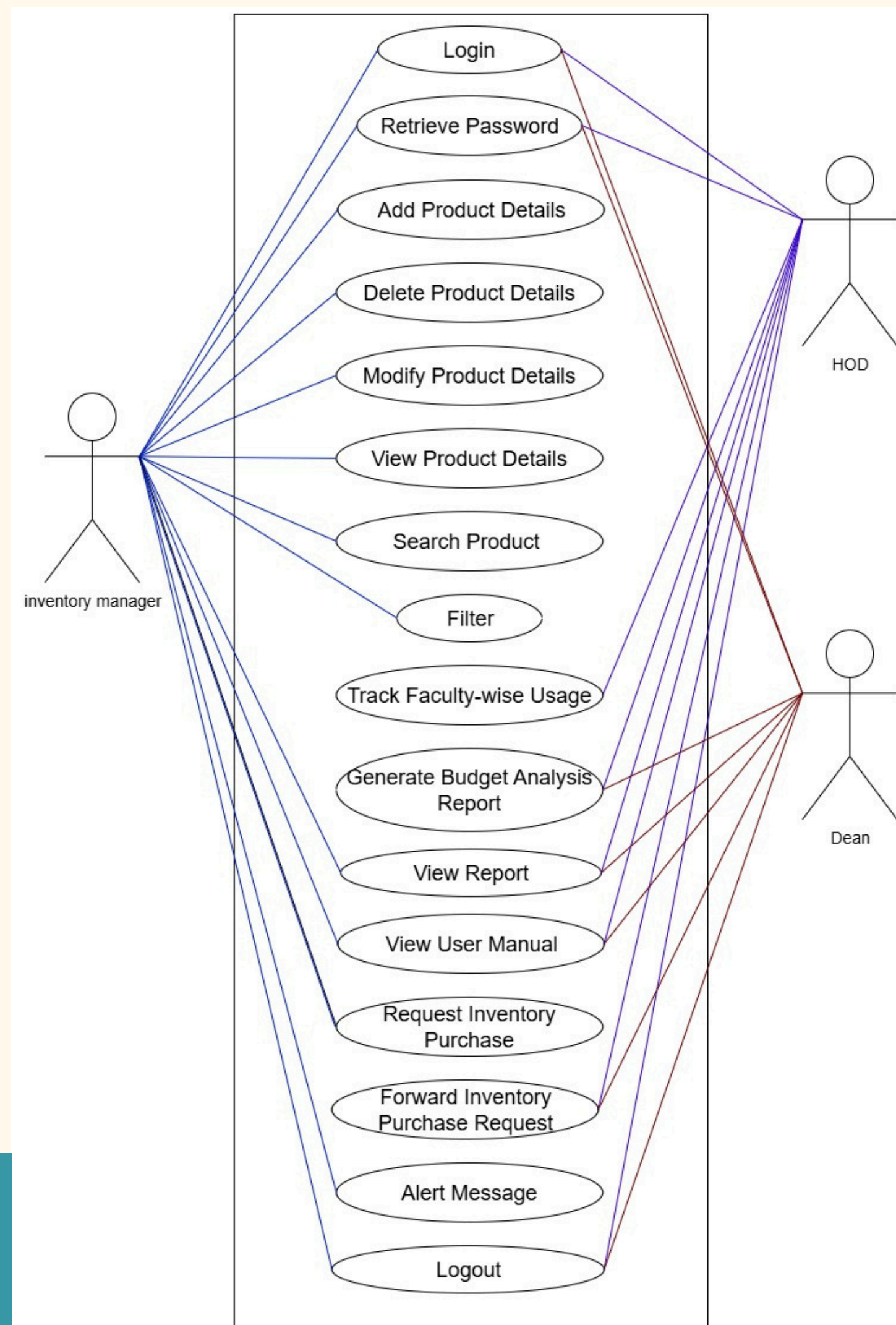


SOFTWARE AND HARDWARE REQUIREMENTS

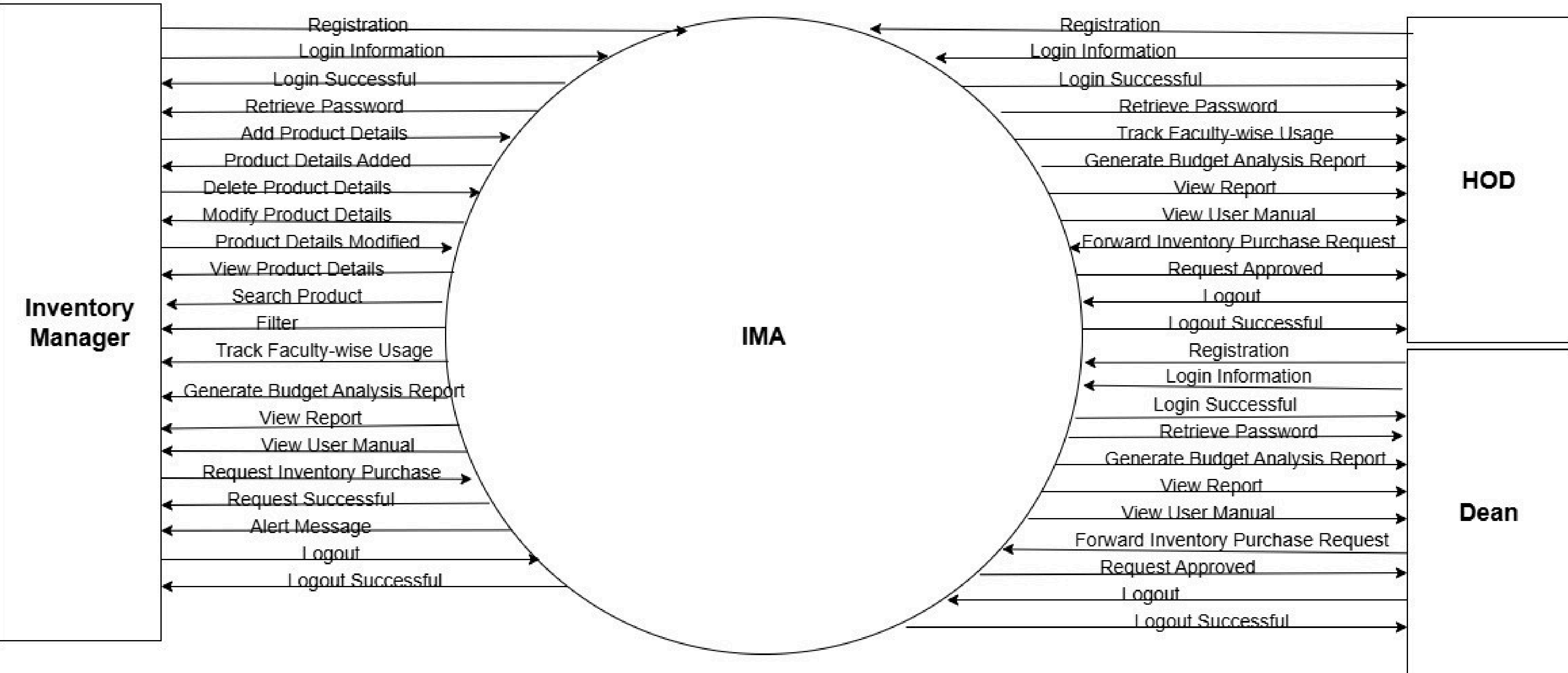


Use-Case diagram:

- Inventory Managers: End users who manage procurement and allocation through the application, while Inventory Managers also receive automatically generated monthly inventory reports on their dashboard to support both daily management and periodic operational reviews.
- HOD and Dean(Administrator): Occasional user overseeing reports and handling administrative monitoring along with request approvals.

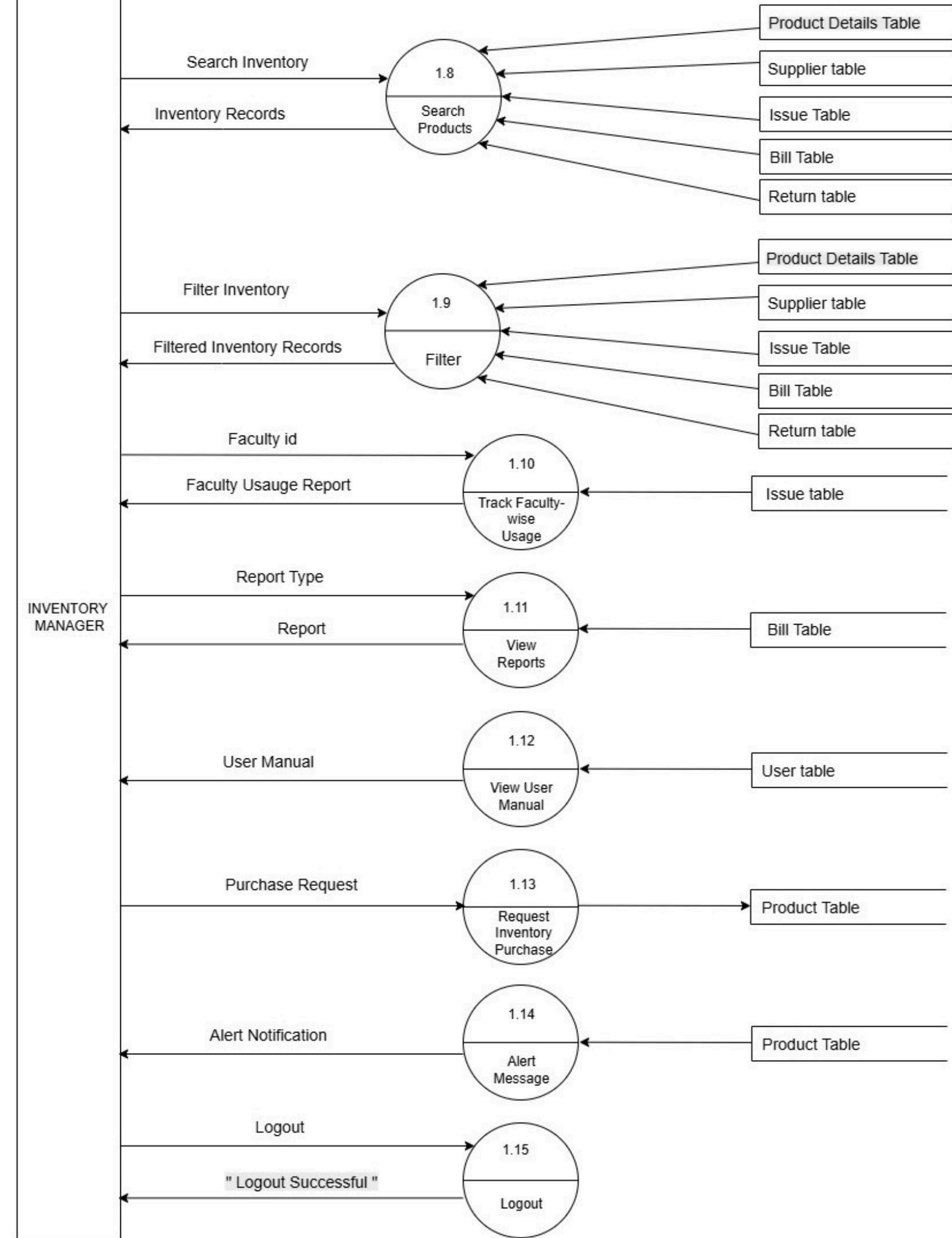
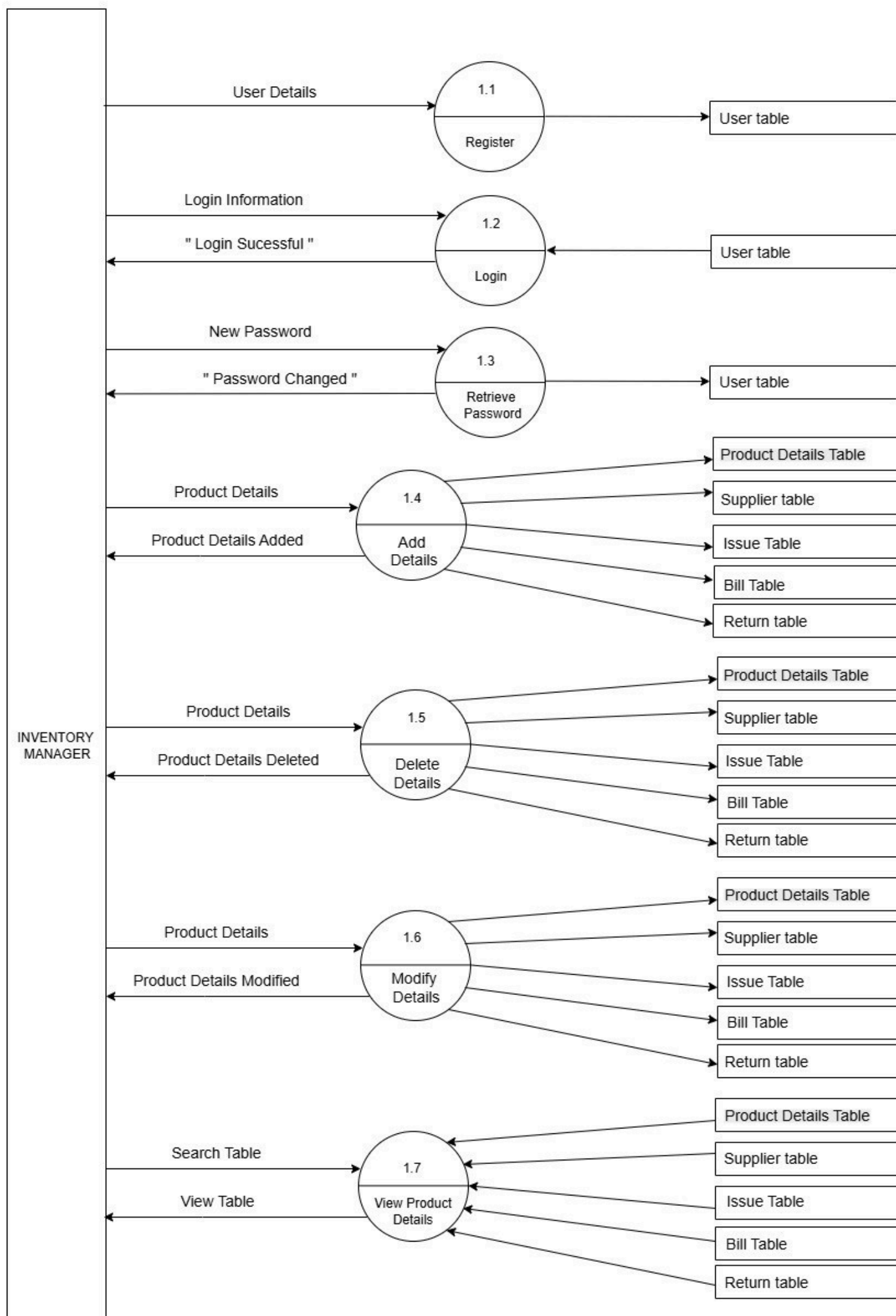


DFD LEVEL-0 :



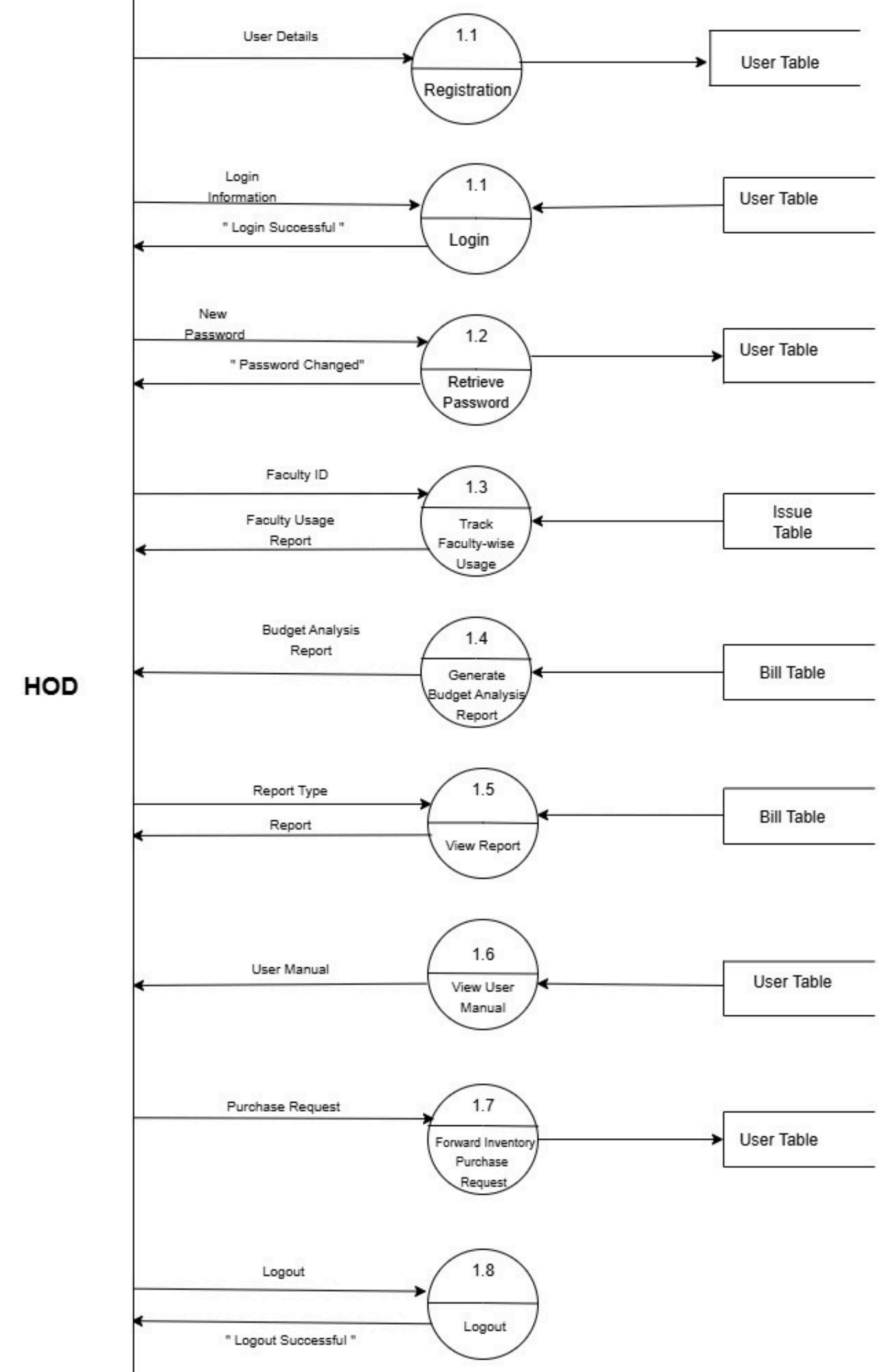
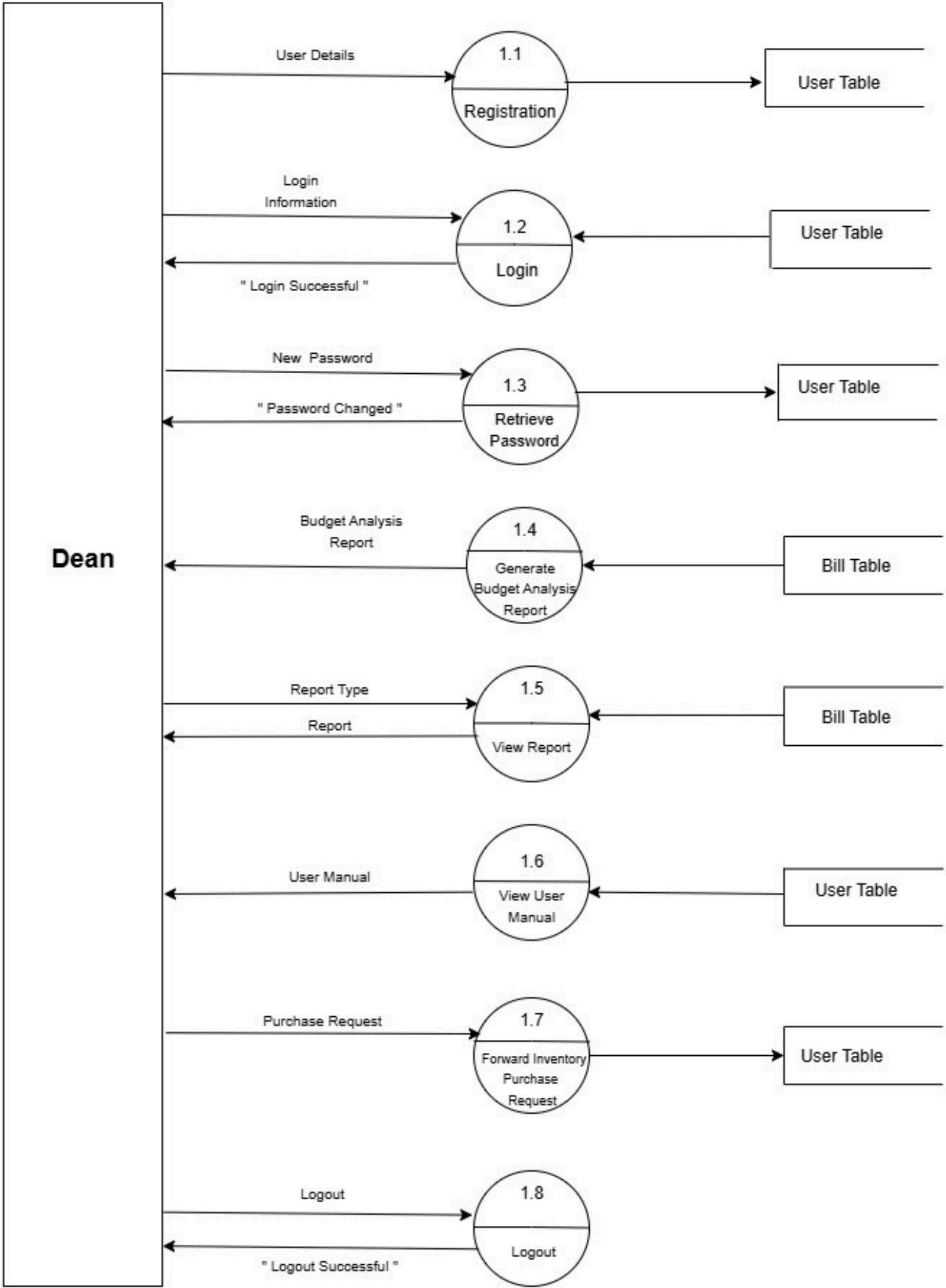
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LEVEL-1:

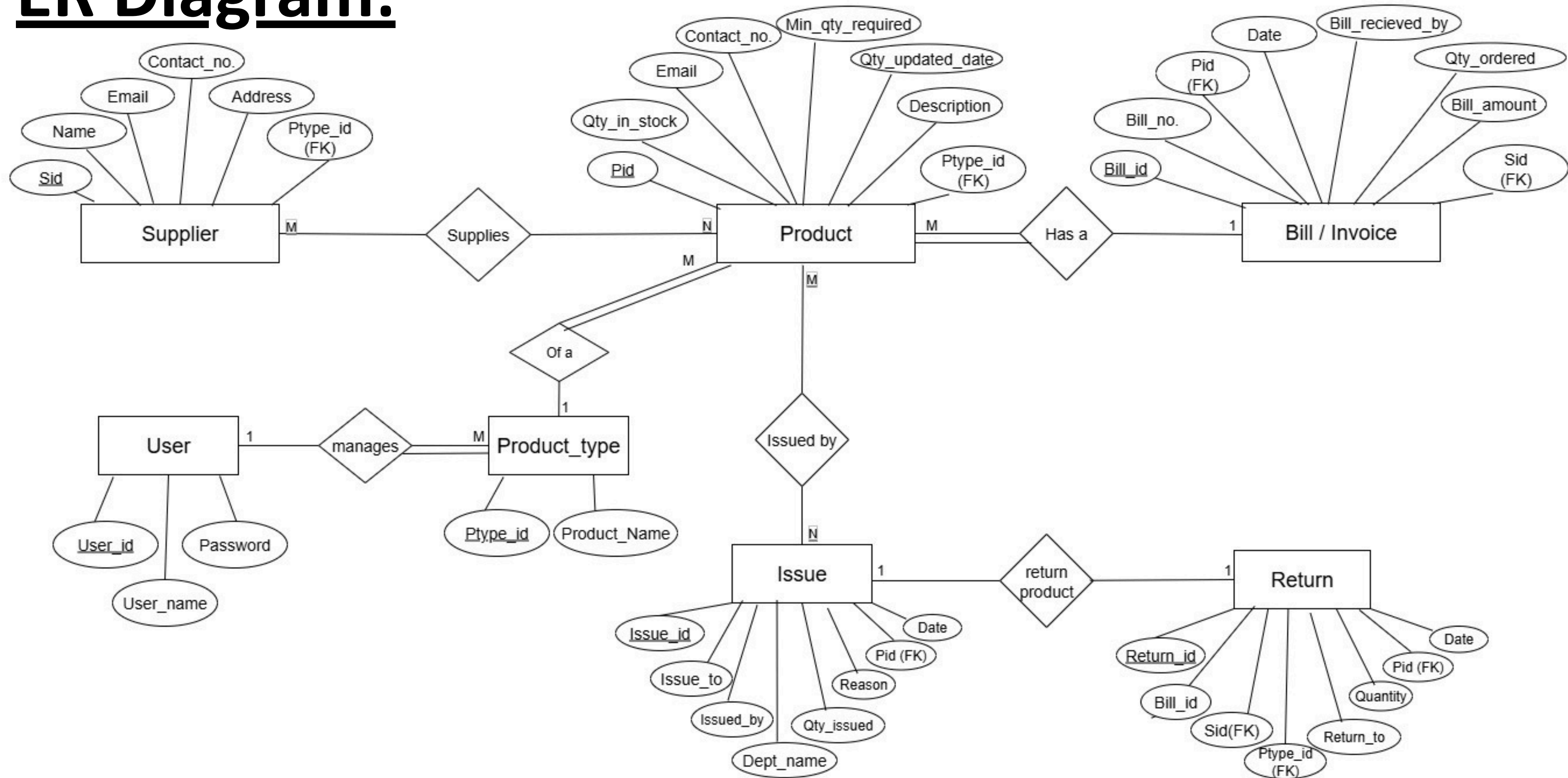


DFD

LEVEL-1:



ER Diagram:



User Interfaces

INVENTORY MANAGEMENT APPLICATION

USERNAME :

PASSWORD :

[FORGOT PASSWORD?](#)

LOGIN

User Manual **Logout**

Update Issue Return

View Report Budget Analysis

Track Usage Request Inventory

Supplier Name		Bill	UPLOAD 
Email		Bill Recieved by	
Address		Date Of Purchase	
Contact number		Quantity Ordered	
Product Type ID			
Product Name			
Product Description			
Date Of Update			



THANK YOU
